

Learning from Click-Logs on MIND and EB-NeRD

Assignment 2 — Two-stage retrieve-then-rank with behavioural features

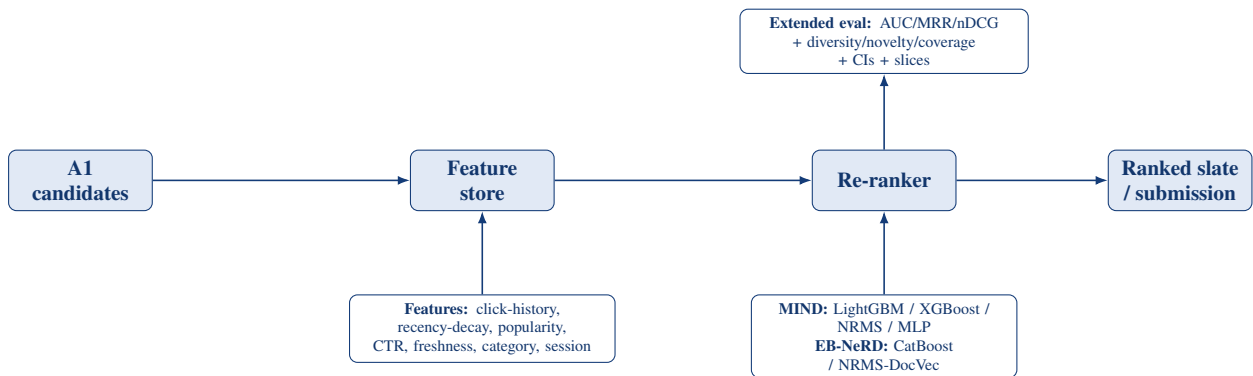
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CS4.406 Information Retrieval & Extraction — Assignment 2

<https://github.com/shivam-git-acc/ClickLogs>

1 What We Built

Building on the Assignment 1 lexical/semantic retrieval pipeline, we add behavioural signals from click-logs and a learned re-ranker, on two datasets — **MIND** (English, Microsoft) and **EB-NeRD** (Danish, Ekstra Bladet). The system is a *two-stage retrieve-then-rank* pipeline: Assignment 1’s candidate generator (BM25 / embedding retrieval, or the impression’s in-view slate) supplies top- K candidates, and a learned re-ranker scores them over engineered behavioural features. Every behavioural feature is computed *point-in-time* (strictly-before- T) under an enforced behaviour-window boundary. Both datasets are trained on the *large* bundles; extended evaluation is on labelled dev; final predictions are submitted to both Codabench leaderboards. Work split: Shivam ran the MIND pipeline end-to-end; Suyash ran the EB-NeRD pipeline end-to-end; the design, feature taxonomy, and evaluation protocol are shared.



Leaderboards (Q5)

MIND test AUC **0.6581** (Codabench 13967, submission 934066), up from A1’s 0.6435. **EB-NeRD** final CatBoost predictions on all 13,536,710 test impressions (205,925,868 candidate rows) validated and packaged for Codabench (comp. 2469).

2 Click-History & Session Features (Q1)

The feature set combines the A1 ranking signal with click-history, article, and session features, all leakage-safe and point-in-time.

MIND (14 features). Mean / best-match / recency-decay-weighted history-embedding similarity (all-MiniLM-L6-v2), click count, rolling popularity (1h/24h/7d/total), CTR (24h/total), freshness (time-since-first-seen), category-match with user history, position, slate size. MIND history carries no timestamps, so the exponential recency weighting decays by *recency rank* (the most-recent clicks weigh highest).

EB-NeRD. A1 hybrid score; log-pageviews / log-inviews; a 24h-half-life *decayed CTR* with Bayesian smoothing toward the global rate ($\alpha=20$); a 10-bucket article-age encoding; semantic-history (cosine

of candidate vs sum-pooled, L2-normalised history vectors over the supplied 768-d contrastive artifact) and BM25-history over the same sparse click matrix; category / subcategory affinity; article sentiment; normalised position; a *causal* zero-based session depth (from EB-NeRD’s `session_id`, positions assigned within each session by impression time). `read_time` and `scroll_percentage` are excluded as post-impression outcomes not known at serving time. Three temporal CTR variants are kept — FRESH (includes first half of validation), STALE (train-only), FINAL (all labels before the test week); the main validation uses STALE to match the train-only regime expected before test.

Behaviour-window boundary (Q9). Every count uses only events strictly before the impression time. MIND enforces this with a per-article `bisect` index plus a leakage assertion (**0 future-click violations** over 5,000 sampled impressions). EB-NeRD keeps train / validation / test histories as separate objects with an executable assertion that they never mix, and derives split boundaries directly from each behaviour table (Table 1).

Table 1: EB-NeRD large temporal splits (one week each; history sourced per split).

split	impressions	interval
train	12,063,890	2023-05-18 07:00 → 05-25 06:59
validation	12,566,385	2023-05-25 07:00 → 06-01 06:59
test	13,536,710	2023-06-01 07:00 → 06-08 06:59

The EB-NeRD catalogue is 125,541 articles; the test set holds 205,925,868 in-view candidate rows (15.2 candidates/impression). The supplied contrastive artifact stores a 768-d vector per article (≈ 386 MB float32 after normalisation). **Training scale:** MIND GBDT is trained on a 600k-impression sample of MINDlarge (22.5M candidate rows; GBDT saturates by $\sim 400k$ per A1), and MIND NRMS on up to the full 2.2M — the title-only baseline diverged at $lr\ 10^{-3}$ (dev AUC 0.585) and the improved variants use $lr\ 10^{-4}$ + gradient clipping on an 800k/2-epoch budget. EB-NeRD CatBoost trains on 300k sampled impressions (3.32M rows).

3 Re-Ranker (Q2)

MIND. Both GBDT options (LightGBM + XGBoost, LambdaMART / pairwise over the 14 features) and both neural options (a canonical NRMS over titles; a small MLP over the engineered features). GBDT trained on a 600k sample of MINDlarge (GBDT saturates well before full scale, per A1’s data-scale finding); NRMS trained on up to the full 2.2M impressions. **EB-NeRD.** A CatBoost `PairLogit` grouped learning-to-rank re-ranker (700 iterations, depth 8, $lr\ 0.08$, L2 5), trained on 300k sampled impressions (3.32M candidate rows, 301k positives); an NRMS-style frozen-document-vector model (768→512→256, 16-head self-attention user encoder) is retained as a neural alternative.

Before vs after re-ranking (Q2.4). Re-ranking lifts every metric sharply over the retrieval-order baseline on MIND (Table 2); XGBoost narrowly edges LightGBM. On EB-NeRD, CatBoost lifts the A1 reference from AUC 0.6806 to 0.7129.

Table 2: MIND before vs after re-ranking (dev); “before” = retrieval order. EB-NeRD CatBoost shown for reference.

metric	before	LightGBM	XGBoost	EB-NeRD CatBoost
AUC	0.6348	0.7099	0.7110	0.7129
MRR	0.3495	0.4141	0.4160	0.4779
nDCG@5	0.3314	0.3941	0.3961	0.5436
nDCG@10	0.3925	0.4556	0.4573	0.5855

4 Baseline Reproduced, Then Beaten (Q3)

4.1 MIND — feature engineering beats the neural baseline

Baseline: a canonical NRMS (title-only, negative-sampling softmax) reproduced at dev AUC **0.5846**. We beat it two ways (Option C), each with a paired impression-level bootstrap 95% CI on the per-impression AUC delta (Table 3). We also ran two further neural modifications purely as ablations, so the neural improvement is isolated cleanly.

Table 3: MIND Q3: baseline, improvements, ablations (dev AUC) with paired bootstrap 95% CIs.

model	AUC	paired 95% CI (vs stated ref)
NRMS title-only (baseline)	0.5846	—
NRMS +recency (ablation)	0.6383	vs +abstract: -0.0206 [-0.023 , -0.019] (<i>regression</i>)
NRMS +category (ablation)	0.6572	vs +abstract: -0.0017 [-0.004 , $+0.0004$] (<i>null</i>)
NRMS title+abstract (Improv 2)	0.6589	vs baseline: $+0.0743$ [$+0.071$, $+0.078$] sig.
MLP (features)	0.6863	—
LightGBM (Improv 1)	0.7099	vs baseline: $+0.1212$ [$+0.111$, $+0.131$] sig.
XGBoost	0.7110	—

Headline — inputs beat architecture, and the ablations are honest

Both improvements clear the baseline with CIs excluding zero. The neural ablation is instructive: the *abstract* helps ($+0.074$, sig.); *category* is a proven null (CI spans zero — the text already encodes topic); *recency-weighting* is a proven regression (-0.021) — list-position is a noisy signal without real timestamps. The MLP (0.686, neural) beats NRMS (0.659, neural) not by architecture but by *inputs*: NRMS reads only text, while the MLP sees the behavioural features (CTR, popularity, recency) that drive MIND clicks. Feature-based GBDT (0.710) beats every neural variant.

Top LightGBM gains confirm the story: `ctr_total`, `hist_emb_mean`, `ctr_24h`, `hist_emb_best`, then `slate_size / cat_match / freshness`. Behavioural CTR dominates; content similarity is a secondary but real contributor on English MIND.

4.2 EB-NeRD — CatBoost beats the A1 hybrid

Baseline: the A1 hybrid at AUC 0.68060 (tuned blend pv 3.0 / ctr 0.5 / age 3.0 / sem 1.0 / bm25 0.5 / sub 0.5). **Improvement:** CatBoost PairLogit at **0.71287**, $+0.03227$ AUC, improving all four metrics with paired bootstrap 95% CIs excluding zero (Table 4). The NRMS-DocVec neural alternative reached only 0.53547 (below the reference), so CatBoost is the final EB-NeRD re-ranker — an empirical result for this protocol, not a general claim that GBDT dominates neural rankers.

Table 4: EB-NeRD Q3: CatBoost vs A1 reference, paired bootstrap 95% CI (1,000 resamples, 60k impressions).

metric	A1 ref	CatBoost	Δ & 95% CI
AUC	0.68060	0.71287	$+0.03227$ [0.03016, 0.03435]
MRR	0.45311	0.47789	$+0.02478$ [0.02211, 0.02761]
nDCG@5	0.50802	0.54361	$+0.03559$ [0.03298, 0.03839]
nDCG@10	0.55911	0.58548	$+0.02637$ [0.02411, 0.02867]

4.3 Feature-contribution ablations (both datasets)

EB-NeRD history ablation. Masking the personalization path (semantic/BM25 history, category/subcategory affinity, history length) on the trained CatBoost model isolates the personalization contribution (inference masking, not retraining):

History features contribute measurable within-slate ordering signal (larger MRR/nDCG than AUC drop). The pageviews column is the Q9 serving-safety diagnostic (§8).

EB-NeRD dominant feature — article age. Bucketed article age is the strongest single signal (AUC 0.6706); as a linear $-age$ term it scores 0.4950 (below random), because the click-rate is non-monotone

Table 5: EB-NeRD ablations (full minus masked), paired bootstrap 95% CI.

metric	Δ history	95% CI	Δ pageviews	95% CI
AUC	+0.00699	[.0057, .0082]	+0.04011	[.0382, .0422]
MRR	+0.01082	[.0091, .0127]	+0.04520	[.0426, .0478]
nDCG@5	+0.00862	[.0070, .0102]	+0.05282	[.0505, .0553]
nDCG@10	+0.00839	[.0070, .0098]	+0.03940	[.0372, .0414]

— it peaks at 8–12h and decays both sides ($9.4 \times$ max/min spread across ten buckets): <1h 0.084, 1–2h 0.116, 2–4h 0.126, 4–8h 0.129, **8–12h 0.132**, 12–18h 0.098, 18–24h 0.050, 24–48h 0.033, 48–120h 0.029, >120h 0.014. A monotone recency term cannot represent this, which is why bucketing matters. Other EB-NeRD single-feature AUCs: total_pageviews 0.612, decayed CTR 0.573, category affinity 0.558, semantic 0.540, BM25 0.520.

MIND shows the mirror image: content retrieval is weak but behavioural CTR/popularity carry the reranker (top LightGBM gain `ctr_total`), and history-length slicing (§5) confirms personalization value. **Both teams independently found that freshness/recency is the dominant article-side signal and behavioural CTR the dominant interaction signal.**

Neural alternatives, in detail. MIND NRMS is the canonical title encoder (multi-head self-attention + additive attention, negative-sampling softmax). EB-NeRD NRMS-DocVec projects the frozen 768-d contrastive vectors $768 \rightarrow 512 \rightarrow 256$, applies a 16-head self-attention user encoder with additive attention, and scores by user–article dot product (≈ 0.84 M params, 1M sampled impressions $\rightarrow$ 492k usable, 8 negatives, 8 epochs, $1r\ 10^{-4}$). Both neural rankers under-performed their respective feature-based re-rankers on these protocols.

5 Extended Evaluation & Slices (Q5)

All metrics carry bootstrap 95% CIs; both required slices (cold/warm, head/tail) are reported.

MIND per-reranker $\times$ slice. Evaluating all four rerankers across identical slices (Table 6) yields two non-obvious findings: (i) **LightGBM wins even on cold-start users** — behavioural signal is largely *candidate-side* (an article’s CTR, popularity, freshness need no user history), so content models do not rescue cold-start; (ii) **the MLP beats LightGBM on head articles (0.752 vs 0.744) but collapses on the tail (0.621)**, whereas LightGBM stays balanced (tail 0.677) — its overall win comes from tail robustness, not head accuracy, because trees carve feature-space regions for rare articles while the MLP over-weights popularity.

Table 6: MIND per-reranker AUC by slice (dev). Cold/warm = history-length p20/p80; head/tail = clicked-article popularity.

model	all	cold	warm	head	tail
NRMS title-only	0.5846	0.5727	0.5897	0.6028	0.5671
NRMS title+abstract	0.6589	0.6323	0.6664	0.6837	0.6352
MLP	0.6863	0.6498	0.6942	0.7521	0.6213
LightGBM	0.7099	0.6765	0.7174	0.7437	0.6766

EB-NeRD extended eval. On the 60k STALE validation subset the CatBoost model gives category diversity@10 0.7761, novelty@10 37.07, and coverage@10 0.0194 ($\approx 1.94\%$ of the 125,541-article catalogue). Slice AUCs (Table 7): the tail slice ($n=58,596$) scores 0.7215, higher than the small head slice ($n=835$, AUC 0.386 — interpret cautiously). No cold-start metric is reported because the selected subset contained zero impressions under the <5-history-click definition, reported honestly rather than fabricated.

6 Two-Stage Retrieval at Scale & Serving (Q2, Q4)

Table 7: EB-NeRD slices (60k STALE validation, CatBoost).

slice	n	AUC	MRR	nDCG@5	nDCG@10
warm (all 60k)	60000	0.71287	0.47789	0.54361	0.58548
head	835	0.38629	—	—	—
tail	58596	0.72154	0.48347	0.55063	—
cold	0	no impressions under <5-click definition			

ANN retrieval (MIND, full 104k pool). We built FAISS indexes over all article embeddings and compared exact vs approximate search: HNSW is **15 $\times$ faster** than exact Flat (p50 1.01ms vs 14.73ms; p99 1.47ms vs 23.21ms) for 18% more memory (188 vs 160 MB) — the ANN scale-up A1 only theorised. Content retrieval recall@200 is low (0.021); we tested two candidate-augmentation strategies (global-CTR, category-personalized) and *both reduced recall* (0.017, 0.018): on MIND the clicked article is user- and moment-specific, so popularity heuristics displace useful content candidates. On the retrieved set, LightGBM dominates the rerankers (conditional MRR 0.222 vs NRMS 0.076 vs MLP 0.031) — trees are robust to the retrieval distribution shift; the MLP is brittle to it.

Serving latency (Q4, measured). MIND end-to-end p99 is well under a 100ms SLA (Table 8); **feature-build dominates** (72% of latency), the first optimisation target (hot-article feature caching). EB-NeRD’s CatBoost re-rank stage measures p50 33.2ms / p99 38.4ms per impression (model 2.95 MB; 386 MB dense vector store).

Table 8: Serving latency: MIND full pipeline (1,000 requests) and EB-NeRD re-rank stage.

stage (MIND)	mean	p99	EB-NeRD	value
retrieval (HNSW)	1.13ms	1.50ms	CatBoost p50	33.2ms
feature build	11.80ms	22.18ms	CatBoost p95	35.4ms
LightGBM rerank	3.45ms	5.34ms	CatBoost p99	38.4ms
NRMS rerank	5.41ms	7.13ms	model size	2.95 MB
E2E (LightGBM)	16.38ms	27.14ms	vector store	386 MB

Cost / QPS. MIND LightGBM path: ≈ 488 QPS on an 8-core VM, $\approx \$0.17$ per 1000 queries. EB-NeRD re-rank stage: latency-implied serial capacity ≈ 26 req/s per core (back-of-envelope; a monetary figure needs a fixed instance price).

7 Where It Breaks at 10 $\times$ (Q4)

10 $\times$ load (same catalogue). The app tier is *stateless*, so it scales horizontally behind a load balancer with flat p99 — until the **feature store** saturates first: point-in-time popularity/CTR via `bisect` over growing event lists dominates at 10 $\times$ QPS, so production needs a streaming windowed-counter store (e.g. Redis) rather than re-scanning event lists, plus a hot-article feature cache. **10 $\times$ catalogue.** The **dense vector store** is the binding constraint (MIND 160 MB $\rightarrow$ 1.6 GB; EB-NeRD 386 MB $\rightarrow$ 3.86 GB, $O(Nd)$), and exact Flat search degrades $O(N)$ — HNSW/IVF (already validated) becomes mandatory, with product quantization and dimensionality reduction as pre-ANN options (the 768 dimensions are supplied, not chosen). **Scoring** is the largest absolute cost but least concerning: batches are independent, so it distributes across cores/hosts with no shared state (EB-NeRD’s 206M-row test pass runs batch-parallel; MIND’s per-request scoring is $O(\text{slate})$). **Neural path** (NRMS / NRMS-DocVec) is GPU-bound and costlier per request, so the GBDT/CatBoost path is the production choice on cost and latency.

8 Anti-Gaming & Correctness (Q9)

Both datasets enforce the behaviour-window boundary with an executable leakage test (MIND: 0 future-click violations; EB-NeRD: split-history separation assertion). We report metrics with and *without* serving-unavailable features. On EB-NeRD, `total_pageviews` (whose 7-day aggregation can post-date the impression for articles published in the test week) is future-contaminated: masking it drops AUC

by $+0.04011$ $[0.03823, 0.04218]$, quantifying the model’s dependence on it — reported as a masking diagnostic, not the performance of an independently retrained serving-safe model. History-masking and pageview-masking both use the trained model at inference. Metrics are computed within-impression (never pooled across slates of differing length; an apparent slate-position effect resolves to $1.00\times$ within fixed-length slates), and the official competition scorer is vendored into the test suite (AUC exact; rank metrics exact on distinct scores). Final test submissions are re-parsed line-by-line to verify ID order, slate length, and permutation validity before archiving.

Leaderboard Submissions (Q5)

MIND: test AUC **0.6581** (Codabench 13967, submission 934066), up from A1’s 0.6435; the dev→test drop (0.710→0.658) is the expected concept-drift tax documented in A1. **EB-NeRD:** the final CatBoost model scores the complete test candidate slate in bounded batches — 13,536,710 impressions and 205,925,868 candidate rows. The prediction file (one ordered permutation per impression) was validated line-by-line for count, impression-ID order, slate length, and permutation validity, then packaged as a ≈ 230 MB ZIP for Codabench comp. 2469. (*Screenshots below.*)

ID # ▾	File name	Date	Status	Score	Actions
934066	results1.zip	2026-09-20 02:06	Finished	0.6581	  
897651	mind_minilm_submission.zip	2026-08-23 00:33	Finished	0.6435	  
897338	mind_large_submission.zip	2026-08-22 19:06	Finished	0.6292	  
895472	mind_submission_full.zip	2026-08-20 22:41	Finished	0.6145	  
894392	mind_submission.zip	2026-08-19 23:59	Finished	0.5877	  

ID # ▾	File name	Date	Status	Score	Detailed Results	Actions
932319	ebnerd_a2_catboost_predictions.zip	2026-09-18 22:22	Submitted			
931764	ebnerd_nrms_predictions.zip	2026-09-18 16:40	Submitted			
901313	ebnerd_predictions.zip	2026-08-25 17:50	Finished	0.6817		 
899780	ebnerd_predictions.zip	2026-08-24 18:22	Submitted			

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Figure 1: Codabench leaderboard submissions for MIND (top) and EB-NeRD (bottom).

References. [1] CS4.406 IRE Assignment 2 specification, 2026. [2] Kruse et al., EB-NeRD: A Large-Scale Dataset for News Recommendation, RecSys Challenge 2024. [3] Wu et al., Neural News Recommendation with Multi-Head Self-Attention (NRMS), EMNLP-IJCNLP 2019. [4] Wu et al., MIND: A Large-Scale Dataset for News Recommendation, ACL 2020.